ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #26

DeskCraft v0.1.0

Support tickets built from e-mail for a small business: threading, business-hours SLA, reply drafts as .eml, macros, escalation to taskcraft and unhappy-customer signals to gaincraft. One SQLite file, one binary that is both CLI and MCP server. Part of the back-office craft suite (replaces Zendesk / Freshdesk / a shared support inbox).



v0.1.0	3 Crates	3527	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\deskcraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

Support tickets built from e-mail for a small business: threading, business-hours SLA, reply drafts as `.eml`, macros, escalation to taskcraft and unhappy-customer signals to gaincraft. One SQLite file, one binary that is both CLI and MCP server. Part of the back-office craft suite (replaces Zendesk / Freshdesk / a shared support inbox).

Status: **v0.1.0, Tier 1** (store + CLI + MCP, tested). No desktop window yet. Synthetic demo mail only. **Nothing is ever sent by deskcraft**: replies are `.eml` drafts you send from your mail client.

- How to use it day to day: HELP.md
- What it is for and where it stops: docs/PRD.md
- Every MCP tool: docs/TOOLS.md (generated, checked by a test)
- Changes: CHANGELOG.md

What it does

- **Ingest** `.eml` files or a maildir (`cur/`, `new/`), offline. MIME, quoted-printable, base64 and RFC 2047 headers are decoded by `mail-parser`. Re-ingesting the same folder is a no-op (Message-ID and content hash are unique). `--dry-run` shows what would happen.
- **Threading** in this order: Message-ID / In-Reply-To / References, then the `[D-12]` tag deskcraft puts into reply subjects, then same requester + same subject (Re/AW/WG stripped) within 30 days. Mail from your support address counts as your reply and stops the first-response clock. A reply to a closed ticket opens a new ticket with a note.
- **Tickets:** status new / open / pending / solved / closed, priority low..urgent ("dringend", "urgent" in the subject raise it), assignee, tags, internal notes, full history with actor.
- **SLA** in business hours (Mon-Fri 08:00-17:00, Austrian public holidays excluded): first response and resolution per priority, states met / ok / at_risk / breached with due times.
- **Replies:** `.eml` drafts with correct In-Reply-To / References, UTF-8 subject encoding, signature; macros fill `{{first_name}}`, `{{ticket}}`, `{{subject}}` and can set a status and tags.
- **Escalation:** `ticket.escalated` to taskcraft (becomes an issue there), once per ticket.
- **Signals:** deterministic DE/EN keyword heuristic (unzufrieden, kündigen, Frechheit, refund, ...) finds unhappy customers; `--emit` sends `signal.negative` to gaincraft, once per ticket.

Quick start

```
deskcraft demo-seed
deskcraft list
deskcraft show D-1
deskcraft sla
deskcraft reply D-2 --macro geloest
deskcraft escalate D-3 --project OPS
deskcraft signals
```

Your own mail: `deskcraft config --address support@yourfirm.at --name "Support"`, export the support folder as `.eml` (Outlook: drag mails into a folder; Thunderbird: ImportExportTools NG) and `deskcraft ingest D:\Mail\Support`.

Database: `--db PATH`, else `$DESKCRAFT_DB`, else `%LOCALAPPDATA%\deskcraft\deskcraft.db`. Outbox: `--outbox`, else `outbox` next to the database.

MCP

```
{ "mcpServers": { "deskcraft": { "command": "D:/Dev/cargo-
target/release/deskcraft.exe", "args": ["mcp", "--allow-path", "D:/Mail/Support"] } } }
```

14 tools, all `desk_*`; see docs/TOOLS.md. Agent actions are recorded with actor `agent`. Without `--allow-path` the server cannot read any mail folder.

Build

```
just ci           # clippy -D warnings, tests (incl. docs/TOOLS.md sync), fmt check
just docs         # regenerate docs/TOOLS.md after changing a tool
just build-release
```

Needs `../craft-biz` checked out next to this repo.

Not done (honest list)

See the "Gaps" section of docs/PRD.md. In short: no window, no IMAP, no sending, pending does not pause the SLA clock, no attachments stored.

Product Requirements & Architecture Specification

What it is

A local-first support desk for a 1-10 person business: customer e-mails become tickets, replies are drafted with threading intact, service levels are measured in business hours, and problems flow on to engineering (taskcraft) and account health (gaincraft). Pure Rust, ArtCraft pattern: one engine, CLI + MCP today, egui later.

Part of the back-office craft suite: `mcp-central-docs/projects/craft-fleet-expansion/BACKOFFICE_CRAFT_SUITE.md`, section 4.6.

Users

- The person answering support mail (often the owner). Uses the CLI today, the egui app later.
- Agents over MCP: triage (priority, tags, assignee), draft replies from macros, watch the SLA, escalate. Every agent action is in the ticket history with actor `agent`.

Jobs

1. Turn a folder of support mail into a deduplicated, threaded ticket list.
2. Know what is waiting and what is about to breach.
3. Answer consistently (macros) without losing thread headers.
4. Hand bugs to engineering and unhappy customers to account management without copy-paste.

Today (v0.1.0)

- Ingest `.eml` / maildir with MIME decoding; idempotent; dry run; per-file errors.
- Threading by headers, `[D-n]` tag, then subject + requester within 30 days; own replies recognised by support address.
- Tickets with status, priority, assignee, tags, notes, history.
- Business-hours SLA (AT holidays from craft-biz) with due times and at-risk warning.
- Reply drafts as RFC 5322 `.eml` (CRLF, UTF-8, RFC 2047 subject, In-Reply-To, References), macros with placeholders, mark-as-sent.
- Bus events: `ticket.escalated` -> taskcraft (verified end to end), `signal.negative` -> gaincraft (queued until gaincraft exists).
- 14 MCP tools (docs/TOOLS.md, generated and test-checked), CLI with tables and a conversation view.

Boundaries

- **No sending.** deskcraft never talks SMTP. Sending stays in the user's mail client until SMTP credentials can come from Windows Credential Manager (craft-biz secrets, not built).
- **No mailbox login.** IMAP only after checking whether lookcraft's local store can be the source, so the fleet has one mail stack, not two.
- **No AI sentiment.** The signal is a documented keyword heuristic. LLM triage may come later through craft-ai-core and will be labelled advisory.
- **No customer portal, no chat, no phone.** E-mail only.
- **Personal data:** mail bodies are stored locally in SQLite. No telemetry, no cloud.

Gaps (honest)

- No egui window.
- Pending does not pause the SLA clock; SLA targets and business hours are constants, not settings.
- Attachments are not stored (body text only; HTML-only mail is converted to text).
- Subject fallback compares case-insensitively for ASCII only (SQLite `lower`); "Büro" vs "BÜRO" would not match.
- Outlook `.msg` files are not supported; export as `.eml`.
- No merge/split of tickets; no customer/organisation records beyond the e-mail domain.
- Demo data only; never run against a real support mailbox yet.

Next

1. egui app: ticket list with SLA colours, conversation view, reply composer with macro picker.

2. SLA settings (hours, targets) and pause on pending.

3. lookcraft store as an ingest source; then SMTP send behind `--send` with Credential Manager.

Complete MCP Tool & CLI Verb Reference

Generated by `deskcraft tools --markdown`. Do not edit by hand: a test fails when this file is out of date. Every tool is also a CLI verb (`deskcraft --help`). Dotted names (`app.verb`) are accepted.

Tool	What it does	Required	Optional
`desk_status`	Ticket counts by status, messages, unsent drafts, support address, outbox folder.	-	-
`desk_config`	Show or set the support address (mails from it count as our replies), the sender name and the reply signature.	-	`signature`, `support_address`, `support_name`
`desk_ingest`	Read .eml files or a maildir (cur/new) into tickets. Threads by Message-ID / In-Reply-To / References, then [D-n] subject tags, then same requester + subject within 30 days. Re-ingesting is a no-op. dry_run stores nothing.	`path`	`dry_run`
`desk_ticket_list`	Tickets, urgent first. Default: new, open and pending.	-	`assignee`, `include_closed`, `priority`, `status`, `text`
`desk_ticket_show`	One ticket with all messages, internal notes, history and SLA clocks.	`ticket`	-
`desk_ticket_update`	Change status, priority, assignee (null clears), tags (replaces the set) or add an internal note.	`ticket`	`assignee`, `note`, `priority`, `status`, `tags`
`desk_reply_draft`	Write a reply as an .eml draft into the outbox (threaded headers, [D-n] in the subject). Nothing is sent. body or macro; {{name}}, {{first_name}}, {{ticket}}, {{subject}}, {{agent}} are filled.	`ticket`	`body`, `macro`
`desk_reply_mark_sent`	Mark a reply draft as sent (after sending it from your mail client). The first sent reply stops the first-response clock.	`message`	`at`
`desk_macro_set`	Create or change a reply macro, optionally setting a status and adding tags when used.	`name`, `body`	`add_tags`, `set_status`
`desk_macro_list`	Reply macros.	-	-
`desk_sla_status`	First-response and resolution clocks of open tickets in business hours (Mon-Fri 08:00-17:00, AT holidays excluded): met, ok, at_risk, breached.	-	`as_of`
`desk_escalate`	Send a ticket to taskcraft as ticket.escalated (becomes an issue there).	`ticket`	`details`, `project`

Tool	What it does	Required	Optional
	Once per ticket; tags it escalated.		
`desk_signals`	Tickets whose customer mails read as unhappy (deterministic DE/EN keywords, not AI). emit=true sends signal.negative to gaincraft once per ticket.	-	`emit`
`desk_demo_seed`	Fill an EMPTY desk with labelled demo mails (MIME encoded, threaded, one angry customer) and macros.	-	-